Learning Discriminative Embedding for Hyperspectral Image Clustering Based on Set-to-Set and Sample-to-Sample Distances

Yao Qin, *Student Member, IEEE*, Lorenzo Bruzzone, *Fellow, IEEE*, and Biao Li

***Abstract*— Recently, deep learning techniques have been introduced to address hyperspectral image (HSI) classification problems and have achieved the state-of-the-art performances. In this article, we propose a novel clustering algorithm for HSI based on learning embedding using the set-to-set and sample-to-sample distances (LSSDs). This technique consists of four main components: 1) oversegmentation; 2) generation of set-to-set and sample-to-sample distances; 3) learning embedding by training a siamese network; and 4) density-based spectral clustering. First, the HSI is oversegmented into superpixels by using the entropy rate superpixel (ERS) algorithm. Second, the set-to-set distances are obtained by representing the segmented sets of samples as affine hull (AH) models, whereas the sample-to-sample distances are computed by employing the local covariance matrix representation (LCMR) method. Third, sample pairs with the smallest and largest similarities are extracted according to the two distances. Then, these pairs are fed into the siamese multilayer perceptron (MLP) network and discriminative embeddings are learned by training the network with contrastive loss. Finally, density-based spectral clustering is applied to the deep embedding to obtain clustering results. Experimental results on three real HSIs demonstrate that the proposed method can achieve better performance than the considered baseline methods.**

***Index Terms*— Affine hull (AH), clustering, distance learning, hyperspectral image (HSI), local covariance matrix representation (LCMR), remote sensing, sample-to-sample distance, set-to-set distance, siamese network.**

I. Introduction

HYPERSPECTRAL images (HSIs) capture contiguous spectral information on a wide range of the electromagnetic spectrum [1]–[3] and, therefore, are suitable for distinguishing different materials, facilitating various remote sensing applications, such as environmental monitoring [4], mineral exploration [5], and land-cover classification [6]–[8]. One fundamental challenge in these applications is HSI classification, which is usually addressed in the framework of supervised learning [9]. Recently, several state-of-the-art performances have been achieved by deep learning techniques [10]. However, the training of supervised deep learning requires a very huge number of labeled samples. In this article, we focus on unsupervised learning based on clustering. Ideally, clustering assigns similar samples of HSI to a group that corresponds to one specific class without requiring any labeled sample. However, in general, HSI clustering is a difficult and challenging task due to the large spectral variability caused by multipath scattering, nonhomogeneous mixing of pixel-level spectra, and variations in sun-canopy-sensor geometry [11].

According to the machine-learning literature, clustering methods are divided into five categories based on their working mechanisms [12]: 1) *centroid-based*; 2) *density-based*; 3) *biological-based*; 4) *spectral-based*; and 5) *deep-learning-based* methods. Among the centroid-based clustering methods, k-means [13] and fuzzy c-means (FCM) [14] are the two typical techniques widely used that group pixels by iteratively updating clustering centroids. Their variations include k-means++ algorithm [15], FCM with spatial constraint (FCM_S) algorithm [16], and deep k-means [17] that jointly learns the data points' representations and their cluster assignments. In the literature of density-based clustering, classical methods are the clustering by fast search and find of density peaks (FSDP) algorithm [18] and its improved version [19]. The FSDP algorithm is based on the idea that clustering centers are characterized by both a higher density than their neighbors and a relatively large distance from points with higher densities. The spectral-based clustering methods are highly popular algorithms, which work by embedding the data in the eigenspaces of the Laplacian matrices and applying k-means on this representation to obtain the clusters [12]. The recently developed sparse subspace clustering (SSC) method [20] applies the spectral-based clustering after obtaining the self-representation coefficient matrix. In addition, several deep learning-based clustering methods have been recently proposed, including the deep embedded clustering (DEC) [21] and the information maximizing self-augmented training (IMSAT) techniques [22]. As a pioneer work, the DEC method simultaneously learns feature representations and cluster assignments using deep neural networks, whereas the IMSAT method combines fully connected network and regularized information maximization (RIM) [23], which learns a probabilistic classifier such that mutual information between inputs and cluster assignments is maximized. The

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state-of-the-art clustering performance on the well-known MNIST pattern recognition data set [24] is achieved by the IMSAT method.

When these methods are directly applied to HSI, it is difficult to achieve satisfactory clustering performance as most of them consider spectral information as the only feature and, thus, the information deficiency caused by the spectral features affects the accuracy. Indeed, several deep learning methods have been proposed to extract spectral–spatial features and, thus, achieve state-of-the-art classification performance, such as the squeeze multibias network in [25] and the deformable CNN-based HSI classification method in [26]. Similarly, several spectral–spatial clustering methods have been proposed, exploiting both the spectral and spatial features to obtain better clustering performances. Typical techniques include FCM_S and Markov random fields (MRF)-based clustering algorithm [27]. Based on the SSC method, spatial weighted SSC (SwSSC), SSC incorporating spatial information (SSC_S), spatial sparse subspace clustering (S^4SSC) [28], and l_2-norm regularized SSC (L2SSC) [29] have been proposed for HSI clustering. Recently, a novel total variation (TV) regularized collaborative representation clustering with a locally adaptive dictionary algorithm for HSI has been proposed in [30] and a spectral–spatial diffusion learning (DLSS) method that combines a geometric estimation of class modes with a diffusion-inspired labeling that incorporates both spectral and spatial information has been proposed in [31].

The above-mentioned methods based on sparse subspace, representation model, and diffusion learning (DL) proved to be more accurate than many other clustering methods as they exploit the spatial information of HSI. However, there are two main issues associated with these methods. First, since both the SSC and the collaborative representation model include the representation coefficient matrix, clustering of large HSI may be prohibitive due to the high computational cost of the representation coefficient matrix. For example, it may take nearly 3 h to obtain a clustering map of 100×200 pixels by the L2SSC method. Second, since the DLSS method combines density-based techniques with geometric learning through diffusion geometry, it is difficult to obtain accurate clustering if the samples within clusters have large variations, whereas the deep learning-based methods are more robust to spectral variations than the conventional methods for HSI classification.

To address these problems, we propose a deep learning clustering method of HSI by learning the discriminative embedding based on the set-to-set and sample-to-sample distances (LSSDs). To the best of our knowledge, we are the first to address the issue of applying deep learning to HSI clustering. Specifically, the proposed method consists of four main components: 1) HSI oversegmentation; 2) generation of set-to-set and sample-to-sample distances; 3) learning discriminative embedding by training a siamese network; and 4) density-based spectral clustering. The siamese architecture is employed in this article as it can work in an unsupervised manner and can learn the intrinsic structure of the samples [32]. Intuitively, discriminative features can be obtained by using (dis)similar pairs to train the siamese network. However, it is difficult to generate accurate (dis)similar pairs of HSI samples by simply using Euclidean distance and spectral angle measure (SAM) as samples show strong spectral variabilities. To address this problem, two distance measures denoted as set-to-set and sample-to-sample distances are employed to generate (dis)similar sample pairs. These two distance measures based on the affine hull (AH) model [33] and the local variance matrix representation (LCMR) method [34] are jointly employed to achieve pairs generation. This is done for the following two reasons. First, compared with the SAM and the Euclidean distances, both of them exploit the similarities and variances of local samples and proved to be suitable for HSI classification. Indeed, the accuracy of generated pairs has direct effect on the learning of discriminative embedding, and these two distances show the state-of-the-art performance in measuring distance between HSI samples. Second, they are based on different mechanisms and work at different levels (set and sample), thus obtaining complementary results. Indeed, the siamese network requires rich and accurate (dis)similar pairs to capture the intrinsic structure of the samples, and fusing these two distances can generate richer pairs to achieve more accurate clustering results.

To be more specific, oversegmentation is first achieved by applying the entropy rate superpixel (ERS) algorithm and the superpixel number is estimated by detecting the continuous edges of a binary edge map derived from the first principal component of the HSI. Second, the samples in each superpixel are considered as a set and then set-to-set distances are computed by treating each set as an AH model. In addition, the sample-to-sample distances are derived from the LCMR method. Third, given the set-to-set and sample-to-sample distances, the (dis)similar pairs of HSI samples are extracted on the basis of the largest and smallest distances. To better exploit the spatial information for training the network, joint spectral–spatial features derived from the first four principal components are also extracted. Then, a siamese network of multilayer perceptron (MLP) with contrastive loss is trained by using these sample pairs of spectral–spatial features. Finally, when the deep embedding of HSI samples is achieved, the FSDP-based spectral clustering method is applied to the learned embeddings to achieve final clustering. The rationale of using the FSDP-based spectral clustering method is as follows: 1) the embedded features learned by the siamese network may present as nonconvex clusters, whereas spectral clustering can easily capture the nonconvex characteristic of these clusters; 2) although the embedding of the samples belonging to similar classes (such as "asphalt" and "bitumen" classes) may overlap, the siamese network makes the learned embedding more discriminative and denser, indicating that it is feasible to employ the density-based FSDP clustering method to discover the true clustering centers. Therefore, the FSDP-based spectral clustering facilitates achieving accurate clustering maps. Comprehensive experiments on three benchmark HSIs have been conducted to demonstrate the effectiveness of the proposed algorithm.

The rest of this article is organized as follows. The AH model-based distance and the LCMR algorithm are reviewed

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in Section II. The proposed methodology of HSI clustering is presented in Section III. Section IV describes the experimental setup and results. Section V summarizes the contributions of our research.

II. Background Algorithms

This section briefly describes the background algorithms, i.e., the AH model-based distance and the LCMR algorithm, which play a relevant role in the proposed method.

A. AH Model-Based Distance

In mathematics, the AH model of a set $\mathbf{S}$ in Euclidean space is the smallest affine set including $\mathbf{S}$. For better illustration, two sets consisting of multiple samples are denoted as $\mathbf{S}_1$ and $\mathbf{S}_2$, respectively. The AH model for each set considers each sample in the set as one sample point in a linear or affine space and characterizes the set by a convex geometric region spanned by its samples [33], [35]. Specifically, the AH model can be formulated as

$$\begin{aligned} \mathrm{AH}(\mathbf{S}_i) &= \left\{ \mathbf{x}_i = \sum_j \alpha_{ij}\mathbf{x}_{ij} \middle| \sum_j \alpha_{ij} = 1 \right\} \\ &\mathbf{s.t.}\ \alpha_{ij} \in (0,1), \quad i = 1,2 \end{aligned} \tag{1}$$

where $\mathbf{x}_{ij}$ is the jth sample in set $\mathbf{S}_i$ and α_{ij} is the corresponding representation coefficient. The AH model implicitly treats any affine combination of samples within the set as a valid descriptor of the set and it gives a rather loose approximation to the sample.

To better represent the AH model, any sample μ_s can be chosen as reference and the model is rewritten as

$$\mathrm{AH}(\mathbf{S}_i) = \{\mathbf{x}_i = \mu_{s_i} + \mathbf{U}_{s_i}\mathbf{v}_{s_i}\} \tag{2}$$

where μ_{s_i} can be one of the samples $\mathbf{x}_{ij}$ or the mean of all samples in $\mathbf{S}_i$, $\mathbf{U}_{s_i}$ is an orthonormal basis for the directions spanned by the affine subspace, and $\mathbf{v}_{s_i}$ is a vector of free parameters used to compute the reduced coordinates for the samples within the set and expressed with respect to the basis $\mathbf{U}_{s_i}$ [35]. Generally, the basis $\mathbf{U}_{s_i}$ can be obtained by applying the singular value decomposition (SVD) to $[\mathbf{x}_{i1} - \mu_{s_i}, \cdots, \mathbf{x}_{in_i} - \mu_{s_i}]$, where n_i is number of samples in the set $\mathbf{S}_i$.

Let us consider the two set $\mathbf{S}_1$ and $\mathbf{S}_2$, which are represented as AH models $\{\mathbf{U}_{s1}\mathbf{v}_{s_1} + \mu_{s_1}\}$ and $\{\mathbf{U}_{s2}\mathbf{v}_{s_2} + \mu_{s_2}\}$, respectively. The closest samples between the two AH models can be identified by solving the following minimization problem:

$$\min_{\mathbf{v}_{s_1},\mathbf{v}_{s_2}} ||\{\mathbf{U}_{s1}\mathbf{v}_{s_1} + \mu_{s_1}\} - \{\mathbf{U}_{s2}\mathbf{v}_{s_2} + \mu_{s_2}\}||^2. \tag{3}$$

By defining $\mathbf{U} \equiv (\mathbf{U}_{s_1} \ -\mathbf{U}_{s_2})$ and $\mathbf{v} \equiv \binom{\mathbf{v}_{s1}}{\mathbf{v}_{s_1}}$, we rewrite the problem as the following least squares problem:

$$\min_{\mathbf{v}} ||\{\mathbf{U}\mathbf{v} - (\mu_{s_2} - \mu_{s_1})\}||^2. \tag{4}$$

The optimal $\mathbf{v}^*$ is denoted as $(\mathbf{U}^{\mathrm{T}}\mathbf{U})^{-1}\mathbf{U}^{\mathrm{T}}(\mu_{s_2} - \mu_{s_1})$. Then, the distance between the two sets is defined as[1]

$$\begin{aligned} \mathrm{Dis}_{\mathrm{AH}}(\mathbf{S}_1,\mathbf{S}_2) &= \mathrm{Dis}(\mathrm{AH}(\mathbf{S}_1),\mathrm{AH}(\mathbf{S}_2)) \\ &= ||(\mathbf{I} - \mathbf{P})(\mu_{s_2} - \mu_{s_1})||^2 \end{aligned} \tag{5}$$

where $\mathbf{P} = \mathbf{U}(\mathbf{U}^{\mathrm{T}}\mathbf{U})^{-1}\mathbf{U}^{\mathrm{T}}$ is the orthogonal projection onto the joint span of the directions contained in the two subspaces and $\mathbf{I}$ is the identity matrix.

B. LCMR Method

The LCMR method is inspired by the covariance matrix representation (CMR) method that has been successfully applied to various computer vision tasks, such as image segmentation [36], texture classification [37], and face recognition [38]. Given a set $\mathbf{P}$ consisting of samples $\mathbf{x}_i$ ($i = 1, ..., N$), the CMR is formulated as follows:

$$\mathbf{C}_{\mathbf{P}} = \frac{1}{N-1}\sum_{i=1}^{N}(\mathbf{x}_i - \mu_{\mathrm{P}})(\mathbf{x}_i - \mu_{\mathrm{P}})^{\mathrm{T}} \tag{6}$$

where μ_{P} denotes the mean of all samples. The diagonal entries of the $\mathbf{C}_{\mathbf{P}}$ represent the variance of each sample, whereas the nondiagonal entries represent the covariance among different samples. As covariance matrices (CMs) are symmetric positive definite, they lie on a Riemannian manifold and, thus, Euclidean-based distance is not suitable for them. The Log-Euclidean distance (LED) is proposed in [39] to model the differences of CMs. Given two sets $\mathbf{P}_1$ and $\mathbf{P}_2$, the corresponding CMs are denoted as $\mathbf{C}_{\mathbf{P}_1}$ and $\mathbf{C}_{\mathbf{P}_2}$, respectively. The LED between the two matrices is then defined as

$$d_{\mathrm{LED}}(\mathbf{C}_{\mathbf{P}_1},\mathbf{C}_{\mathbf{P}_2}) = ||\mathrm{logm}(\mathbf{C}_{\mathbf{P}_1}) - \mathrm{logm}(\mathbf{C}_{\mathbf{P}_2})||_{\mathrm{F}} \tag{7}$$

where logm and $||\cdot||$ denote the matrix logarithm operator and Frobenius norm, respectively.

Based on the CMR method and LED metric, the LCMR method[2] exploits the spectral correlation among different bands and spatial correlation in HSI, consisting of three main steps, i.e., maximum noise fraction (MNF)-based dimensionality reduction, similar neighboring pixels extraction, and LED generation. To be specific, the MNF-based dimensionality reduction method is first applied to the original HSI to reduce the dimensionality and discard noise. Then, for each sample in the dimensionally reduced HSI, the SAM distance measurement is used to find its K-1 most similar neighboring samples within the square window of size T. Next, given the sets consisting of K samples for each central sample, the CMs for all samples in the HSI can be easily obtained and then the distance between samples x_i and x_j is defined as

$$\mathrm{Dis}_{\mathrm{LCMR}}(x_i,x_j) = \frac{0.5 \times \mathbf{var}(y_i - y_j)}{\mathbf{var}(y_i) + \mathbf{var}(y_j)} \tag{8}$$

where **var** denotes the vector variance and $y = vec(\mathrm{logm}(\mathbf{C}_{\mathbf{P}})$ represents the vector of CM logarithm on the set $\mathbf{P}$. With this definition, the distances among all samples in the original HSI can be obtained. The main parameters in the LCMR method are the number of MNF components L, the number of neighboring samples K, and the window size T, which are set to 20, 25, and 220, respectively.

[1] http://mlcv.ogu.edu.tr/softwareimageset.html

[2] https://sites.google.com/site/leyuanfang/home/lcmr_tgrs18

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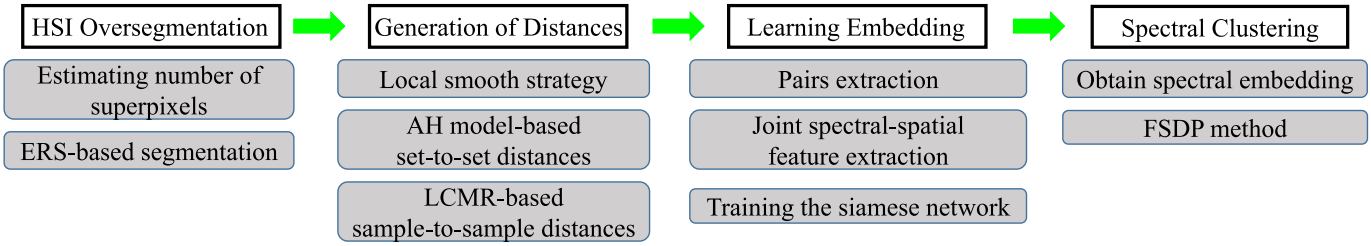


Fig. 1. Illustration of the proposed set-to-set and sample-to-sample distances learning (LSSD) method.

III. Proposed Method

As shown in Fig. 1, the proposed LSSD algorithm consists of four main steps: 1) HSI oversegmentation; 2) generation of set-to-set and sample-to-sample distances; 3) training the siamese network; and 4) density-based spectral clustering. Specifically, since the sets are samples in homogeneous superpixels of HSI, oversegmentation is first applied. It can be divided into two components, i.e., estimating the number of superpixels and segmentation. Second, the generation of the AH model-based set-to-set distances and the LCMR-based sample-to-sample distances is achieved. Once the distances are obtained, Step 3) can be divided into three components, i.e., extraction of positive (similar) and negative (dis-similar) pairs of samples, joint spectral–spatial features extraction, and training of a siamese network with contrastive loss. When the training process finishes, the final clustering can be achieved by employing spectral clustering based on FSDP. In the following, the implementation details of the four main steps are described.

A. Oversegmentation

The details of how to generate superpixels from HSI are given as follows.

1) Estimating the Number of Superpixels: The principal component analysis (PCA) is first used to decompose the HSI and the 2-D map of first principal component $\mathbf{M}_{1pc}$ is obtained. Then, the 2-D binary edge map $\mathbf{M}_{ed}$ is generated by applying the canny detector to the HSI map and the number of continuous edges estimated through the eight-connected component labeling algorithm[3] is considered as the superpixel number N_s.

2) Superpixel Oversegmetation: Given the superpixel number N_s and the 2-D map of the first three principal components $\mathbf{M}_{3pc}$, the oversegmentation is achieved by employing the ERS method [40]. Specifically, the ERS method[4] maps the image $\mathbf{M}_{3pc}$ to a graph $\mathbf{G} = (\mathbf{V}, \mathbf{E})$, where the vertexes $\mathbf{V}$ and the edge weights $\mathbf{E}$ denote the pixels in $\mathbf{M}_{3pc}$ and the pairwise similarities between pixels, respectively. The objective function includes the entropy rate of the random walk on $\mathbf{G}$ and a balancing term encouraging clusters with similar sizes, reducing the number of unbalanced superpixels. When the optimization process of the ERS method finishes, N_s connected subgraphs corresponding to homogeneous superpixels are generated.

[3]https://ww2.mathworks.cn/help/images/ref/bwconncomp.html
[4]https://github.com/mingyuliutw/EntropyRateSuperpixel

B. Generation of Set-to-Set and Sample-to-Sample Distances

Prior to generating both the AH model and the LCMR method-based distances, a local smoothing strategy is introduced to exploit the spatial consistency of HSI. Specifically, given a superpixel consisting of n samples and one of these samples x_i, $\lceil n/20 \rceil$ samples having closest Euclidean distances with regards to sample x_i are extracted, where $\lceil \cdot \rceil$ denotes rounding a number to the next larger integer. Then, the local smoothed HSI is obtained by filtering the sample x_i as the mean of the $\lceil n/20 \rceil$ samples.

Following [33], the AH model-based distances among the sets of oversegmented superpixels are employed. Given the oversegmention results, each sample in the HSI belongs to a unique superpixel and the samples in each superpixel are considered as a set. Then, the distances between these sets are computed by using (5). Note that the sets consisting of less than ten samples are discarded as no enough spectral–spatial information is available to generate accurate set-to-set pairs. Generally, a smaller subset (no more 30 samples) of each set is selected to replace the original one before distance computation. Specifically, no more than 30 sample pairs with high spectral similarity are selected for each set pair and SAM is employed to measure the spectral similarity. Following [34], the LCMR-based distance in (8) is employed to compute sample-to-sample distances.

C. Learning Discriminative Embedding

When the set-to-set and sample-to-sample distances are generated, positive (similar) and negative (dis-similar) sample pairs are first extracted and then discriminative embedding is learned by training a siamese network with these pairs of joint spectral–spatial features. The corresponding details are given as follows.

1) Pairs Extraction: Given the ith sample x_i belonging the set $\mathbf{S}_0$ and the generated distances, the two sets having smallest and largest set-to-set distances from $\mathbf{S}_0$ are denoted as $\mathbf{S}_1$ and $\mathbf{S}_2$, respectively. The samples with the smallest and largest sample-to-sample distances are denoted as x_j and x_k, respectively. Then, the positive and negative pairs related to x_i are extracted considering the following five aspects: 1) since the sets are obtained through oversegmentation, the samples within the set $\mathbf{S}_0$ and x_i are positive pairs; 2) the samples in the similar set $\mathbf{S}_1$ and x_i are positive pairs; 3) (x_j, x_i) is a positive pair; 4) the samples in the set $\mathbf{S}_2$ and x_i are negative pairs; and 5) (x_k, x_i) is a negative pair. In this way, three types of positive pairs and two types of negative pairs can be extracted for each sample in the HSI.

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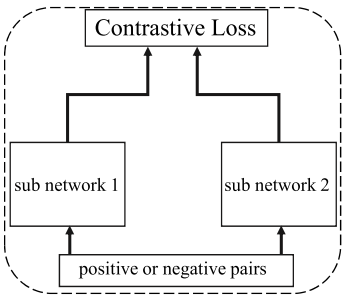


Fig. 2. Illustration of siamese architecture.

2) Joint Spectral-Spatial Feature Extraction: Since the prevailing deep learning methods for HSI classification make use of spatial information to achieve the state-of-the-art performance [41]–[44], the joint spectral–spatial features of HSI are extracted before training the siamese network. To be specific, the first p components in a neighbor region of 3×3 pixels for each sample are flattened and appended to the raw HSI spectral features to form the joint spectral–spatial features. For better illustration, the dimension of the raw spectral features is denoted as d_{spe}. Thus, the dimension of the spectral–spatial feature is $d_{\text{spe}} + 9p$ and the value of p is set to 4 in the proposed method.

3) Training the Siamese Network: Given the extracted positive (negative) sample pairs, we aim to learn a discriminative similarity metric. Since siamese networks are effective for addressing tasks related to finding similarities or relationships between two comparable inputs, we employ a siamese network based on MLP. Fig. 2 illustrates the siamese architecture consisting of two identical subnetworks with the same configuration of parameters and weights, which is typically trained on a collection of positive or negative sample pairs. The siamese architecture is suitable to the clustering task for the following two reasons: 1) sharing weights across subnetworks results in fewer parameters to estimate in the training phase, which, in turn, means less samples required and less risk to overfit and 2) the obtained embedding essentially produces representations of the corresponding samples by using similar models to process similar types of samples.

To be specific, it maps each sample x_i into the discriminative embedding $z_i = G_{\theta_{\text{siamese}}(x_i)}$, where θ_{siamese} denotes the shared parameters of the two subnetworks. The network is typically trained to minimize the following contrastive loss function:

$$\mathrm{L}(x_i, x_j) = \begin{cases} ||z_i - z_j||^2, & \text{if } (x_i, x_j) \text{ is a positive pair} \\ \max(c - ||z_i - z_j||, 0)^2, & \text{otherwise} \end{cases} \tag{9}$$

where c is a margin parameter controlling how far away the dissimilarities should be. One can note that the mapped representations of similar samples obtained by the siamese network are closer, whereas the distances between the mapped representations from dissimilar samples are forced to be margin c. Therefore, the siamese network is trained to learn an adaptive nearest neighbor metric. The dimensionality of the MLP network is set to d-2000-2000-500-C, where d and C are the spectral–spatial sample dimension and the number of clusters, respectively. In the training process, one pair for each type (total five) is selected for each sample. Note that since there are several pairs derived from the set-to-set distances for each sample, only one pair for each type is randomly selected.

Algorithm 1 Proposed Set-to-Set and Sample-to-Sample Distance Learning Method

Input: HSI data, cluster number C, margin c, neighbor number N.
Output: Clustering result.
1: Estimate the number of superpixels.
2: Oversegmentation by using ERS method.
3: Apply local smooth strategy to the HSI.
4: Generation of set-to-set distances according to (5).
5: Generation of sample-to-sample distances according to (8).
6: Generation of positive and negative pairs.
7: Extraction of joint spectral-spatial features using PCA.
8: Train the siamese network using the loss in (9).
9: Forward network to learn embeddings.
10: Obtain spectral embedding using the weight in (10).
11: Apply the FSDP method to the spectral embedding.

D. Density-Based Spectral Clustering

Once the siamese network is trained, each sample x_i is modeled as the discriminative embedding z_i. Although the learned embedding by the siamese network exhibits good separation between samples belonging to different classes, the embedding of samples within class may behave as nonconvex clusters. Therefore, it is difficult to obtain satisfactory results by directly applying the conventional clustering methods to the learned embedding. According to this observation, we employ the spectral clustering to process the embedding as it outperforms other popular clustering algorithms and is capable of handling nonconvex clusters. Specifically, the weight w_{ij} between sample x_i and x_j is defined as

$$w_{ij} = \begin{cases} \exp\left(-\dfrac{||z_i - z_j||^2}{2\sigma^2}\right), & \text{if } x_j \in \mathcal{D}_{x_i}^N \\ 0, & \text{otherwise} \end{cases} \tag{10}$$

where σ represents the median value of all learned distances, $\mathcal{D}_{x_i}^N$ represents the N nearest neighbors of x_i, z_i, and z_j are the embeddings of samples x_i and x_j, respectively. Then, the spectral embedding of samples can be achieved by a simple eigen-decomposition of the Laplacian matrix and the FSDP method can be applied to the spectral embedding. The implementation details of the proposed LSSD method are illustrated in Algorithm 1.

IV. Experimental Results and Discussion

In this section, we first introduce the data sets and the baseline methods (Section IV-A). Then, the results are presented to demonstrate the effectiveness of the proposed method (Section IV-B). All experiments are conducted on a PC with 2.93 GHz Intel i7 CPU and 16.0-GB memory. The network

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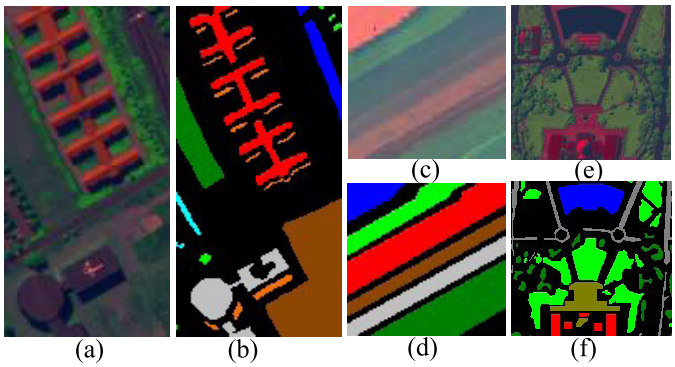


Fig. 3. Data used in our experiments. (a) Color composite image. (b) GT of the Pavia University-A scene. (c) Color composite image. (d) GT of the Salinas-A scene. (e) Color composite image. (f) GT of the Washington DC Mall scene. (Best viewed in color).

training is conducted on a Nvidia Titan Xp GPU with 12-Gb memory.

A. Experimental Setup

1) Data Sets: In order to validate the effectiveness and robustness of the proposed method, three benchmark HSIs are employed. The first data are the HSI collected by the reflective optics spectrographic image system (ROSIS) sensor over the University of Pavia. The Pavia University image[5] contains 103 spectral reflectance bands and has a size of 610×340 pixels. Following [29], [30], a typical subset of 200×100 pixels is employed and eight classes are considered herein: asphalt, meadows, tree, metal sheet, bare soil, bitumen, bricks, and shadows. For better illustration, we denote the subset as Pavia University-A data set. The corresponding composite image and ground truth (GT) are illustrated in Fig. 3(a) and (b). The second data are the Salinas-A image[5] captured by the airborne visible infrared imaging spectrometer (AVRIS) over Salinas Vally, California. As a typical data set for HSI clustering evaluation, Salinas-A image contains 224 bands of 86×83 pixels. Fig. 3(c) and (d) shows the color composite image and the GT, in which six different classes represent mostly different types of crops. The third data are the Washington DC Mall image[6] acquired by the hyperspectral digital imagery collection experiment sensor. Similarly, a subset of 155×160 pixels is used and six main classes are considered: roof, street, grass, tree, water, and square. Fig. 3(e) and (f) shows the corresponding color composite image and the GT. The GT labels of the three data sets are illustrated in Table I.

2) Baseline Methods: Several clustering approaches are employed as baseline methods:

1) **FCM** [14]: FCM[7] is a conventional clustering that is usually employed as baseline.
2) **FCM_S** [16]: FCM_S adopts spatial features in the image segmentation to obtain better clustering performance and its two main parameters are the window size T and spatial weight α, which are set as 3×3 pixels and 0.05 in our experiments, respectively. We reproduce the FCM_S method based on the public available code of FCM[7].
3) **FSDP** [18]: FSDP[8] is a classical density-based clustering method that finds density peaks and is free of parameters. It is applied to the raw HSI samples to demonstrate the merits of the learned embedding over the raw HSI samples.
4) **RIM** [23]: RIM[9] is a framework that simultaneously clusters the data and trains a discriminative classifier. It optimizes an intuitive information-theoretic objective function which balances class separation, class balance, and classifier complexity. The parameter λ is tuned as shown in [23], i.e., we sweep λ across $[0.001, 10]$ and report the best results.
5) **DLSS** [31]: DLSS[10] combines a geometric estimation of class modes with a diffusion inspired labeling that incorporates both spectral and spatial information. It is the state-of-the-art method published on the Salinas-A scene.
6) **DL** [31]: DL[10] learns modes as in DLSS, but the labeling proceeds simply by enforcing that each point has the same spectral labels as its nearest spectral neighbor of higher density.
7) **L2SSC** [29]: L2SSC makes use of the spatial-spectral information contained in HSIs by adding a four-neighborhood two-norm regularizer into the classical SSC model.
8) **TV** [30]: TV applies regularized collaborative representation clustering with a locally adaptive dictionary algorithm for HSI. It is the state-of-the-art method published on the Pavia University-A scene.

TABLE I
NUMBER OF LABELED SAMPLES AVAILABLE FOR PAVIA UNIVERSITY-A, SALINAS-A AND DC MALL DATA SETS

Pavia University		Salinas-A		DC Mall	
Class	GT	Class	GT	Class	GT
Asphalt	425	Brocoli_gw1	391	Roof	575
Meadows	768	Corn_sgw	1343	Street	1639
Tree	63	Lettuce_r4	616	Grass	4036
Metal sheet	1345	Lettuce_r5	1525	Tree	1019
Bare soil	2559	Lettuce_r6	674	Water	1514
Bitumen	860	Lettuce_r7	799	Square	1130
Bricks	94				
Shadow	361				

The typical baseline methods (FCM, FCM_S, FSDP, RIM) are employed for all three data sets. Since it is difficult to reproduce the best results of the other four baseline methods (DL, DLSS, L2SSC, TV) on different data sets, we then compare the proposed LSSD method with the corresponding state-of-the-art methods for different data sets, i.e., L2SSC and TV methods for the Pavia University-A, DL and DLSS methods for Salinas-A and DC Mall data sets. For the FCM, FCM_S1,

[5]http://www.ehu.eus/ccwintco/index.php/Hyperspectral_Remote_Sensing_Scenes
[6]https://engineering.purdue.edu/ biehl/MultiSpec/hyperspectral.html
[7]https://github.com/wwwwwwzj/fcm
[8]https://people.sissa.it/∼ laio/Research/Res_clustering.php
[9]http://www.vision.caltech.edu/∼ gomes/software.html
[10]https://jmurphy.math.tufts.edu/Code/

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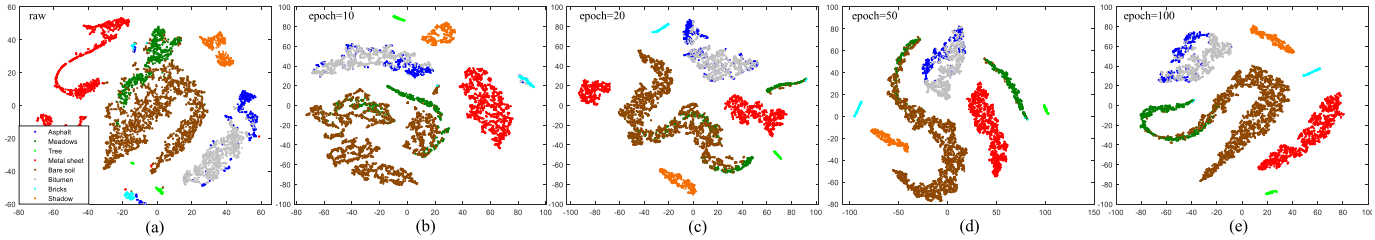


Fig. 4. t-SNE feature visualization of the Pavia University-A data. (a) Raw HSI samples. (b) Learned embedding at epoch=10. (c) Learned embedding at epoch = 20. (d) Learned embedding at epoch = 50. (d) Learned embedding at epoch = 100. (Best viewed in color).

and RIM methods, five trials of the clustering have been performed to ensure the stability of the results, whereas the state-of-the-art results obtained by the DL, DLSS, L2SSC, and TV methods in [29]–[31] are reported. In addition, to validate the effectiveness of the deep embedding, a spectral clustering method based on the set-to-set and sample-to-sample distances (SSD) is employed as a baseline for three data sets. More specifically, if (x_i, x_j) is a positive pair, the weight between x_i and x_j in the SSD method is set as 1, otherwise 0. Once the Laplacian matrix is computed, FSDP is applied to the spectral embedding.

Regarding the proposed LSSD method, we implement the siamese network using the Chainer framework [45], which is an open source deep learning framework written purely in Python language[11]. Inspired by [46], we use the MLP network, where the rectified linear units [47] are used for all hidden activations and batch normalization is applied to each layer to accelerate learning. We set the dimensionality to d-2000-2000-500-C across all the data sets, where d and C are sample dimension and number of clusters, respectively. The RMSprop optimization is used, which keeps the moving average of the squared gradients for each weight and then divides the gradient by square root the mean square [48]. The initial learning rate and batch size are set to 0.0002 and 100, respectively. We divide the learning rate by 10 every 20 epochs and run 100 epochs for each data set. Note that randomly selected three pairs [within set, (dis)similar set], and other two pairs [(dis)similar samples] are used for each sample in the training process. The margin c is set to 5. With regard to the neighbor number N, it is difficult to achieve good performance with too large or too small values: if N is too large, the overlapped samples belonging to different classes may be recognized as one cluster; if N is too small, spectral clustering can hardly capture the distribution of learned embedding and may even recognize one cluster as two. Empirically, the neighbor number N is set to 20 for the three data sets. Note that five trials of training for the siamese network are conducted to obtain stable results.

To numerically evaluate the accuracy of the clustering, the unsupervised overall accuracy (OA), the corresponding average accuracy (AA), and Kappa statistics are used. The OA[12] is defined as

$$\mathrm{OA} = \max_{m} \frac{\sum_{i=1}^{N} \mathbf{1}\{l_i = \mathbf{map}(c_i)\}}{N} \tag{11}$$

[11]https://chainer.org

[12]http://www.cad.zju.edu.cn/home/dengcai/Data/Clustering.html

where **map** denotes the one-to-one permutation between GT labels l and clustering labels c, l_i and c_i are the GT label and clustering assignment produced using the clustering algorithm for x_i, respectively. The permutation **map** is in the range of all possible one-to-one mappings between clustering labels and GT labels. The optimal permutation **map*** can be computed using the Kuhn–Munkres algorithm [49].

B. Results and Discussion

1) Pavia University-A Scene: To validate the effectiveness of the siamese network on the clustering performance, the raw samples and the learned embedding are first compared. Fig. 4 illustrates an example of the t-distributed stochastic neighbor embedding (t-SNE) feature visualization of the Pavia University-A HSI with the raw and learned features. Compared with the raw samples, it is clear that better separation between classes is achieved with the learned features. In addition, the "Tree," "Metal sheet," "Bricks," and "Shadow" classes are well clustered, whereas the other four classes ("Asphalt," "Meadows," "Bare soil," and "Bitumen") show some overlap [see Fig. 4(e)]. Since the t-SNE visualization can only provide qualitative performance, we further compare the clustering performance in terms of the decision graphs in the FSDP, SSD, and the proposed LSSD methods to show the high density performance within classes. Fig. 5 shows the decision graphs with regard to raw HSI samples, spectral embeddings of the SSD and LSSD methods. Indeed, only few samples represent clustering centers with high densities (large ρ values) and large distances from other centers (large δ values) in the ideal discriminative samples. Note that the clustering centers in Fig. 5 are manually selected according to their values of ρ and δ. One can see that the clustering centers in the FSDP and the LSSD methods show better separation than those in the SSD method. However, better clustering performance is achieved by the LSSD method [see Fig. 4(a) and (e)]. The random color map of the oversegmentation results and examples of the clustering maps obtained by different methods is shown in Fig. 6. Clearly, the proposed LSSD method outperforms other baseline methods. One can observe that there is some noise in the clustering maps obtained by the LSSD method [see "Bare soil" in Fig. 6(i)], which may appear inconsistent with the strategy of exploiting superpixels and joint spectral–spatial features in the LSSD method. However, since there is a significant overlapping between samples belonging to the "Bare soil" and "Meadows" classes (see Fig. 4), it is very difficult for the LSSD method to distinguish these samples.

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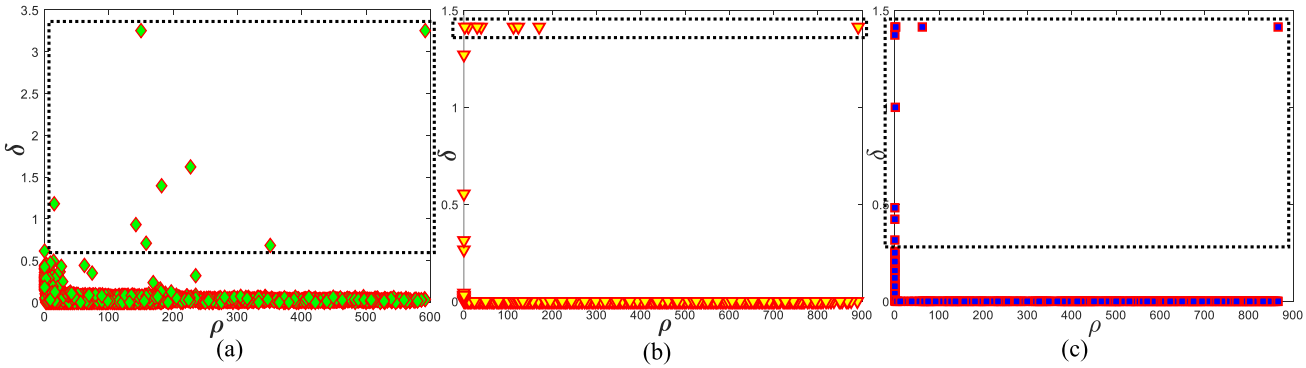


Fig. 5. Illustration of decision graphs for the Pavia University-A data set. (a) FSDP method. (b) SSD method. (c) LSSD method. The eight density centers are marked with black rectangles. (Best viewed in color).

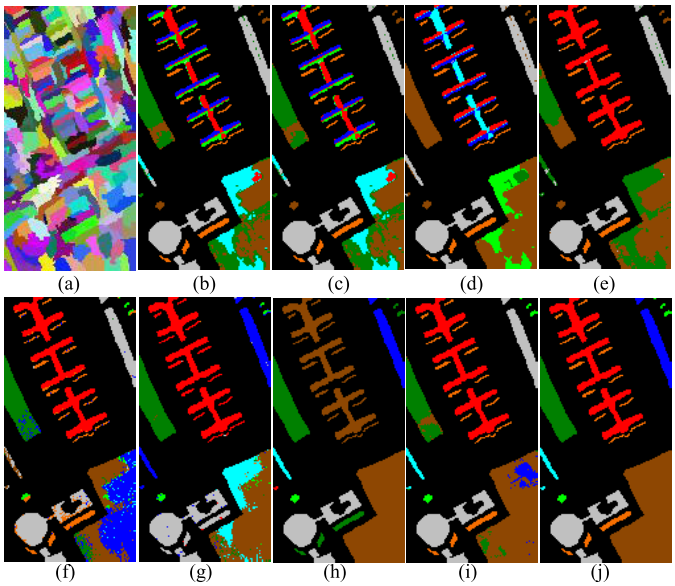


Fig. 6. Oversegmentation result and clustering maps of different methods on the Pavia Unversity-A image. (a) Oversegmentation result. (b) FCM (OA = 54.10%). (c) FCM_S (OA = 54.06%). (d) FSDP (OA = 54.21%). (e) RIM (OA = 68.91%). (f) L2SSC (OA = 63.30%). (g) TV (OA = 79.67%). (h) SSD (OA = 73.33%). (i) LSSD (OA = 87.40%). (j) GT.

This results in salt and pepper noise in the final clustering map for these classes.

Table II reports the mean quantitative evaluation of the clustering results over five trials. It is clear that the FCM, FCM_S, and FSDP methods are inferior to other methods resulting in low mean OAs of 53.99%, 54.06%, and 54.21%, respectively. Compared with FCM_S, the L2SSC method achieves a significantly higher OA (63.30%) by exploiting the spatial contextual information in a much more elaborate perspective. The main improvements of L2SSC lie in the "Tree" and "Metal sheet" classes, with OA increments of 60.32% and 60.00%, respectively. However, larger misclassifications can be easily found in several classes, such as "Bare soil," "Bitumen," and "Bricks." Compared with the proposed LSSD method, the other three baselines (RIM, TV, and SSD) achieve lower mean OAs equal to 66.50%, 79.67%, and 73.33%, respectively. The proposed LSSD method results in the highest mean OA 83.79% with a large improvement on the "Tree" and "Bricks" classes (100% and 97.78%, respectively).

TABLE II

MEAN CLUSTERING ACCURACIES FOR THE PAVIA UNIVERSITY-A DATA SET. THE BEST TWO RESULTS FOR EACH ROW ARE REPORTED IN BOLD AND UNDERLINED, RESPECTIVELY

Class	Methods							
	FCM	*FCM_S*	*FSDP*	*RIM*	*L2SSC*	*TV*	*SSD*	**LSSD**
Asphalt	0	0	0	0	0.71	98.82	**100**	0
Meadows	68.75	82.03	0	63.33	92.97	99.61	**100**	36.74
Tree	0	0	0	12.38	60.32	66.67	49.21	**100**
Metal Sheet	36.30	37.34	37.57	91.97	97.34	**100**	0	**100**
Bare soil	47.71	43.06	69.52	53.07	37.20	68.58	**100**	94.83
Bitumen	**100**	**100**	**100**	98.63	87.67	97.33	**100**	**100**
Bricks	34.04	42.55	0	18.51	0	0	88.30	**97.87**
Shadows	**100**	**100**	**100**	99.45	93.91	0.28	0	**100**
OA(%)	53.99	54.06	54.21	66.50	63.30	79.67	73.33	**83.79**
AA(%)	48.11	49.22	41.37	49.17	56.75	63.72	68.00	**80.80**
Kappa(%)	44.45	45.31	41.51	58.08	56.18	74.59	62.32	**78.16**

2) Salinas-A Scene: Fig. 7 illustrates an example of the t-SNE visualization of the Salinas-A image with the raw and learned features. It is clear that the raw samples overlap across different classes and the learned features at the 100th epoch are well separated. In addition, Fig. 8 shows the corresponding decision graphs related to raw HSI samples, spectral embeddings of the SSD and LSSD methods. One can see that the clustering centers of the raw samples and spectral embedding in the LSSD method show better separation than that in the SSD method. The random color map of the oversegmentation results and examples of the clustering maps obtained by different methods are shown in Fig. 11. It can be seen that the proposed LSSD is superior to other baseline methods with an OA of 99.78%.

To further illustrate the clustering performance, Table III reports the mean quantitative evaluation of the clustering results over five trials. One can observe that the FSDP and RIM methods are inferior to other methods providing mean OAs of 63.00% and 76.01%, respectively. The FCM and FCM_S methods result in OAs of 81.53% and 81.67%, respectively. Compared with FCM_S, the DL and DLSS methods achieve higher OAs equal to 83.13% and 84.76%, respectively.

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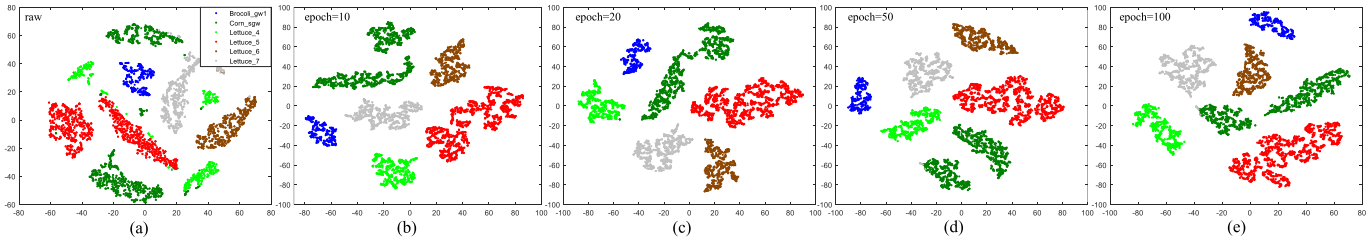


Fig. 7. t-SNE feature visualization of the Salinas-A data. (a) Raw HSI samples. (b) Learned embedding at epoch = 10. (c) Learned embedding at epoch = 20. (d) Learned embedding at epoch = 50. (d) Learned embedding at epoch = 100. (Best viewed in color).

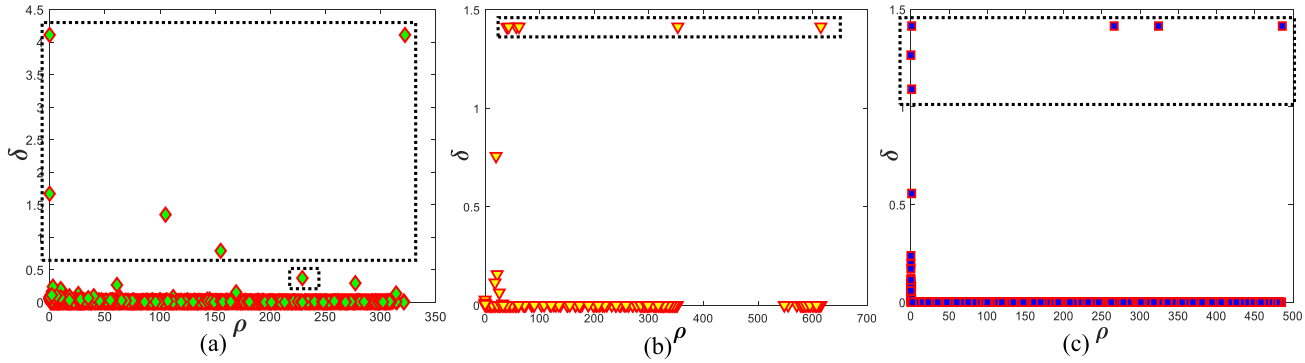


Fig. 8. Illustration of decision graphs for the Salinas-A data set. (a) FSDP method. (b) SSD method. (c) LSSD method. The six density centers are marked with black rectangles. (Best viewed in color).

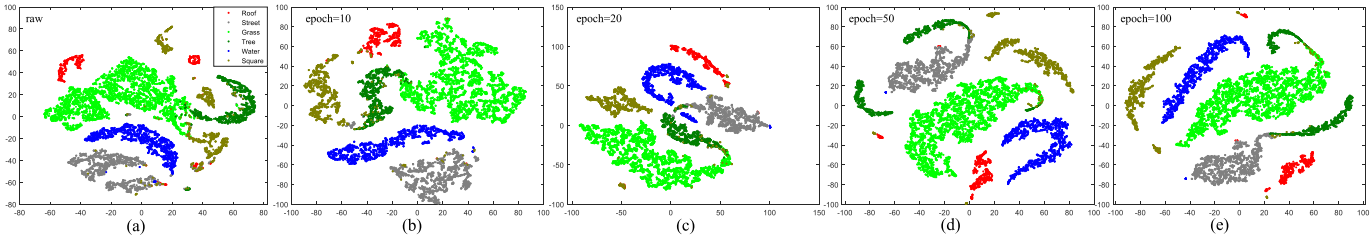


Fig. 9. t-SNE feature visualization of the DC Mall data. (a) Raw HSI samples. (b) Learned embedding at epoch = 10. (c) Learned embedding at epoch = 20. (d) Learned embedding at epoch = 50. (e) Learned embedding at epoch = 100. (Best viewed in color).

Compared with all the other baseline methods, the distance-based methods (SSD and LSSD) result in the highest mean OAs (89.66% and 99.90%) with sharp improvements on the "Corn_sgw" class. One may notice that the proposed LSSD method reaches the state-of-the-art accuracies on almost all classes.

3) DC Mall Scene: Fig. 9 illustrates an example of the t-SNE visualization of the DC Mall HSI with the raw and learned features. It is clear that the raw samples overlap across different classes and the learned features at the 100th epoch are much better in terms of separation across classes. However, there is some overlapping in the learned features, such as the "Roof" and "Street" classes, and the "Tree" and "Grass" classes. Fig. 10 shows the corresponding decision graphs with regard to raw HSI samples, spectral embeddings of the SSD and LSSD methods. One can see that the clustering centers of the raw samples and spectral embedding in the LSSD method show better separation than those in the FSDP and SSD methods. The random color map of oversegmentation result and the corresponding examples of clustering maps obtained by different methods are shown in Fig. 12. It can seen that the highest OA of 97.76% is achieved by LSSD, which is much larger than those obtained by other methods.

To better illustrate the clustering performance, Table IV reports mean accuracies of the clustering results over five trials. One can observe that the SSD and FSDP methods are inferior to other methods providing OAs of 55.70% and 69.60%, respectively. The FCM and FCM_S methods result in similar accuracies, i.e., 86.86% and 86.96%, respectively. Compared with FCM_S, the DL and DLSS methods achieve lower OAs equal to 88.44% and 89.91%, respectively. Compared with the RIM method, which results in a mean OA of 92.84%, the proposed LSSD method achieves a much higher mean OA equal to 97.45%.

4) Distance Effectiveness and Computational Cost: In order to illustrate the accuracies of the oversegmentation and generated distances, we report the following results based on the GT labels: 1) the minimum total number of samples belonging to different classes in all superpixels (oversegmentation errors); 2) the number of similar set pairs in which the two sets

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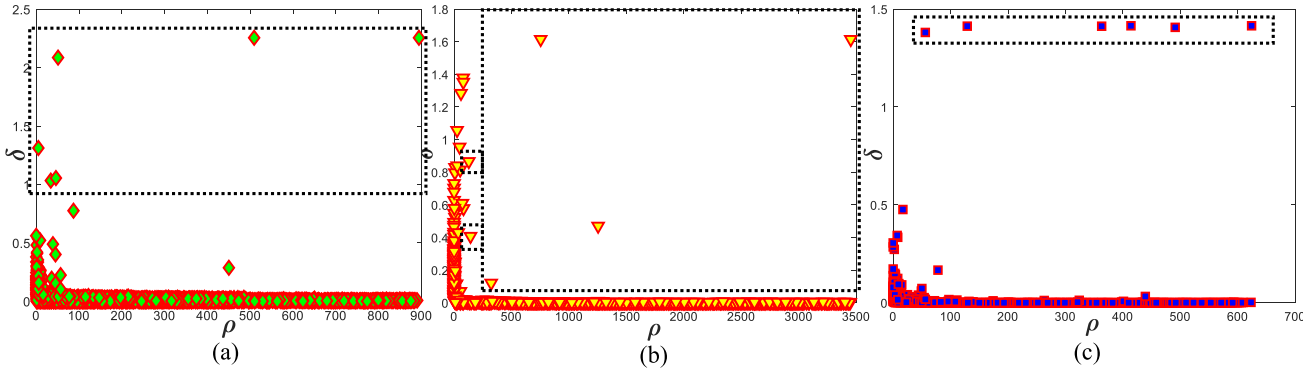


Fig. 10. Illustration of decision graphs for the DC Mall data set. (a) FSDP method. (b) SSD method. (c) LSSD method. The six density centers are marked with black rectangles. (Best viewed in color).

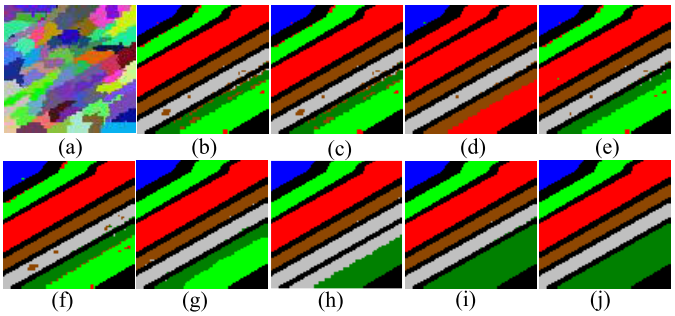


Fig. 11. Oversegmentation result and clustering maps of different methods on the Salinas-A image. (a) Oversegmentation result. (b) FCM (OA = 81.53%). (c) FCM_S (OA = 81.68%). (d) FSDP (OA = 63.00%). (e) RIM (OA = 76.16%). (f) DL (OA = 83.13%). (g) DLSS (OA = 84.76%). (h) SSD (OA = 89.66%). (i) LSSD (OA = 99.78%). (j) GT.

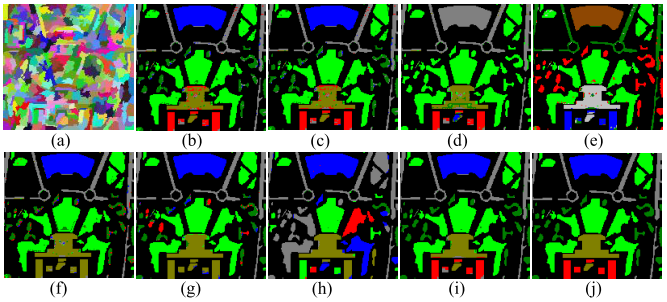


Fig. 12. Oversegmentation result and clustering maps of different methods on the DC Mall image. (a) Oversegmentation result. (b) FCM (OA = 93.43%). (c) FCM_S (OA = 93.62%). (d) FSDP (OA = 69.60%). (e) RIM (OA = 95.84%). (f) DL (OA = 88.44%). (g) DLSS (OA = 89.91%). (h) SSD (OA = 55.70%). (i) LSSD (OA = 97.76%). (j) GT.

TABLE III

MEAN CLUSTERING ACCURACIES FOR THE SALINAS-A DATA SET. THE BEST TWO RESULTS FOR EACH ROW ARE REPORTED IN BOLD AND UNDERLINED. THE PROPOSED METHOD OUTPERFORMS ALL BASELINE METHODS

Class	Methods							
	FCM	*FCM_S*	*FSDP*	*RIM*	*DL*	*DLSS*	*SSD*	**LSSD**
Brocoli_gw1	99.74	99.74	99.49	79.80	99.49	**100**	**100**	**100**
Corn_sgw	31.57	32.29	0	43.22	39.02	40.43	58.97	**100**
Lettuce_r4	95.78	95.29	0	57.76	95.78	**100**	**100**	**100**
Lettuce_r5	**100**	**100**	**100**	89.81	**100**	99.87	**100**	**100**
Lettuce_r6	99.85	99.85	99.70	99.73	**100**	99.11	99.70	99.85
Lettuce_r7	94.87	94.99	98.00	97.00	93.12	99.12	**100**	98.77
OA(%)	81.53	81.67	63.00	76.01	83.13	84.76	89.66	**99.80**
AA(%)	86.65	86.77	51.10	77.38	87.89	90.18	93.18	**99.86**
Kappa(%)	77.49	77.66	52.95	70.02	79.39	81.39	87.22	**99.75**

TABLE IV

MEAN CLUSTERING ACCURACIES FOR THE DC MALL DATA SET. THE BEST TWO RESULTS FOR EACH ROW ARE REPORTED IN BOLD AND UNDERLINED. THE PROPOSED METHOD OUTPERFORMS ALL BASELINE METHODS

Class	Methods							
	FCM	*FCM_S*	*FSDP*	*RIM*	*DL*	*DLSS*	*SSD*	**LSSD**
Roof	90.16	90.30	86.78	70.02	0	0	9.74	**90.61**
Street	93.14	93.24	95.24	94.22	93.96	**98.90**	97.80	97.78
Grass	98.07	98.09	**100**	92.43	96.28	99.48	29.34	98.83
Tree	86.36	86.73	0	97.86	82.43	71.05	15.11	**98.84**
Water	59.45	59.37	0	98.94	98.88	**100**	99.87	99.08
Square	73.19	73.54	71.06	91.26	88.85	91.95	89.65	**92.09**
OA(%)	86.86	86.96	69.60	92.84	88.44	89.91	55.70	**97.45**
AA(%)	82.00	82.19	53.59	90.84	72.20	74.97	59.01	**96.59**
Kappa(%)	82.59	82.72	57.75	90.77	84.74	86.59	45.45	**96.63**

belong to different classes (similar set errors); 3) the number of dissimilar set pairs in which the two sets belong to the same class (dissimilar set errors); 4) the number of similar sample pairs in which the two samples belong to different classes (similar sample errors); and 5) the number of dissimilar sample pairs in which the two samples belong to the same class (dissimilar sample errors). Note that we assume that a set belongs to the same class of the majority of its samples belong to.

Table V shows the OA improvements and error statistics for all the three data sets. For better illustration, the total numbers of samples and sets are also illustrated. For example, the "1/6445" shows that Pavia University-A data set contains 6445 samples and only 1 sample is falsely oversegmented, while "0/232" (third row) means that Pavia University-A data set is divided into 232 sets and all similar set pairs are accurate. One can see that there is only a limited amount of falsely oversegmented samples, validating the effectiveness of the

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TABLE V
OA IMPROVEMENTS (FSDP→LSSD), ERROR STATISTICS OF THE OVERSEGMENTATION AND (DIS)SIMILAR PAIRS, AND COMPUTATIONAL COSTS FOR ALL THE THREE DATA SETS

Datasets	Pavia University-A	Salinas-A	DC Mall
OA improvement (%)	54.21→83.79	63.00→99.80	69.60→97.45
Oversegmentation errors	1/6445	14/5348	88/9913
Similar set errors	0/232	0/71	0/339
Dissimilar set errors	0/232	0/71	2/339
Similar sample errors	4/6445	3/5348	12/9913
Dissimilar sample errors	0/6445	17/5348	84/9913

TABLE VI
COMPUTATIONAL TIME (IN SECONDS) TAKEN BY DIFFERENT METHODS ON THE THREE DATA SETS. THE LARGEST TIMES FOR EACH ROW ARE REPORTED IN ITALIC BOLD

Datasets	Methods							
	FCM	*FCM_S*	*FSDP*	*RIM*	*L2SSC*†	*TV*†	*SSD*	**LSSD**
Pavia-A	2.2	4.7	18.5	197.5	***13315***	6933	70.7	674.6
Datasets	*FCM*	*FCM_S*	*FSDP*	*RIM*	*DL*	*DLSS*	*SSD*	**LSSD**
Salinas-A	4.8	7.2	21.7	134.9	8.0	12.0	71.2	***578.3***
DC Mall	6.3	13.7	47.0	373.8	56.9	99.5	139.9	***1007.9***

† are reported in [29].

oversegmentaion. Compared with sample-to-sample distance, the set-to-set distance is more robust with a lower number of errors over all the three data sets. Although there are false pairs for all the data sets, the proposed LSSD method is robust to learn discriminative embedding, thus increasing sharply the clustering accuracies. Regarding the computational load, the computational times for different methods on the three data sets are reported in Table VI. Clearly, the sparse subspace-based methods (L2SSC and TV) take much larger times than other methods as they include the iterative computation of representation matrices. Compared with the other baseline methods, LSSD takes more time to achieve training with 100 epochs. Note that the time taken from oversegmentation and pairs generation in LSSD ranges between 50 and 100s for the three data sets.

V. CONCLUSION

This article has addressed the issue of HSI clustering in terms of learning discriminative embedding by training a siamese network with (dis)similar sample pairs. Since it is difficult to obtain accurate (dis)similar sample pairs by applying the conventional SAM and Euclidean distances to HSI, the set-to-set distance based on the AH model, and the sample-to-sample distance based on the LCMR method are employed. More specifically, the proposed LSSD method consists of four main components, i.e., HSI oversegmentation, generation of set-to-set and sample-to-sample distances, learning embedding by training a siamese network, and density-based spectral clustering.

The experiments have been conducted on three real HSIs, i.e., Pavia University, Salinas-A, and DC Mall images. The LSSD method achieves much better performance compared with the state-of-the-art clustering methods on all the three data sets. This is mainly due to the following aspects: 1) the AH model-based set-to-set and the LCMR-based sample-to-sample distances both exploit the spectral–spatial similarities and variances of HSI samples; 2) the siamese network is suitable for learning an adaptive metric when the sample pairs are available; and 3) the spectral clustering can well capture the nonconvex clusters of the learned embedding.

The proposed method has two main limitations. First, since it is based on the set-to-set and sample-to-sample distances, it is difficult to obtain satisfied clustering accuracy if these two distance metrics are no longer effective. Second, since the proposed method makes use of the joint spectral–spatial features and requires samples within a superpixel to belong to the same cluster as many other methods based on a similar approach, it is intrinsically not suitable to produce pixelwise scattered clusters. In order to address this limitation, we plan to study more robust distance measures for HSI to obtain accurate pixelwise clustering map as the future development of the proposed LSSD method.

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REFERENCES

[1] M. Fauvel, Y. Tarabalka, J. A. Benediktsson, J. Chanussot, and J. C. Tilton, "Advances in spectral-spatial classification of hyperspectral images," *Proc. IEEE*, vol. 101, no. 3, pp. 652–675, Mar. 2013.
[2] G. Camps-Valls, D. Tuia, L. Bruzzone, and J. A. Benediktsson, "Advances in hyperspectral image classification: Earth monitoring with statistical learning methods," *IEEE Signal Process. Mag.*, vol. 31, no. 1, pp. 45–54, Jan. 2014.
[3] D. Tuia, C. Persello, and L. Bruzzone, "Domain adaptation for the classification of remote sensing data: An overview of recent advances," *IEEE Geosci. Remote Sens. Mag.*, vol. 4, no. 2, pp. 41–57, Jun. 2016.
[4] S. Schneider, R. J. Murphy, and A. Melkumyan, "Evaluating the performance of a new classifier—The GP-OAD: A comparison with existing methods for classifying rock type and mineralogy from hyperspectral imagery," *ISPRS J. Photogramm. Remote Sens.*, vol. 98, pp. 145–156, Dec. 2014.
[5] K. C. Tiwari, M. K. Arora, and D. Singh, "An assessment of independent component analysis for detection of military targets from hyperspectral images," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 13, no. 5, pp. 730–740, Oct. 2011.
[6] F. Melgani and L. Bruzzone, "Classification of hyperspectral remote sensing images with support vector machines," *IEEE Trans. Geosci. Remote Sens.*, vol. 42, no. 8, pp. 1778–1790, Aug. 2004.
[7] C. Persello and L. Bruzzone, "Active learning for domain adaptation in the supervised classification of remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 50, no. 11, pp. 4468–4483, Nov. 2012.
[8] Y. Qin, L. Bruzzone, B. Li, and Y. Ye, "Cross-domain collaborative learning via cluster canonical correlation analysis and random walker for hyperspectral image classification," 2018, *arXiv:1808.09740*. [Online]. Available: https://arxiv.org/abs/1808.09740
[9] L. Bruzzone and C. Persello, "A novel approach to the selection of spatially invariant features for the classification of hyperspectral images with improved generalization capability," *IEEE Trans. Geosci. Remote Sens.*, vol. 47, no. 9, pp. 3180–3191, Sep. 2009.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 02:31:37 UTC from IEEE Xplore. Restrictions apply.

[10] L. Mou, L. Bruzzone, and X. X. Zhu, "Learning spectral-spatial-temporal features via a recurrent convolutional neural network for change detection in multispectral imagery," 2018, *arXiv:1803.02642*. [Online]. Available: https://arxiv.org/abs/1803.02642
[11] D. Lunga, S. Prasad, M. M. Crawford, and O. Ersoy, "Manifold-learning-based feature extraction for classification of hyperspectral data: A review of advances in manifold learning," *IEEE Signal Process. Mag.*, vol. 31, no. 1, pp. 55–66, Jan. 2014.
[12] U. Shaham, K. Stanton, H. Li, B. Nadler, R. Basri, and Y. Kluger, "Spectralnet: Spectral clustering using deep neural networks," 2018, *arXiv:1801.01587*. [Online]. Available: https://arxiv.org/abs/1801.01587
[13] J. Anderson and J. Bodie, "Tree encoding of speech," *IEEE Trans. Inf. Theory*, vol. IT-21, no. 4, pp. 379–387, Jul. 1975.
[14] W. Pedrycz, "Fuzzy sets in pattern recognition: Methodology and methods," *Pattern Recognit.*, vol. 23, nos. 1–2, pp. 121–146, 1990.
[15] D. Arthur and S. Vassilvitskii, "K-means++: The advantages of careful seeding," in *Proc. 18th Annu. ACM-SIAM Symp. Discrete Algorithms*, 2007, pp. 1027–1035.
[16] S. Chen and D. Zhang, "Robust image segmentation using FCM with spatial constraints based on new kernel-induced distance measure," *IEEE Trans. Syst., Man, Cybern. B, Cybern.*, vol. 34, no. 4, pp. 1907–1916, Aug. 2004.
[17] M. M. Fard, T. Thonet, and E. Gaussier, "Deep K-means: Jointly clustering with K-means and learning representations," 2018, *arXiv:1806.10069*. [Online]. Available: https://arxiv.org/abs/1806.10069
[18] A. Rodriguez and A. Laio, "Clustering by fast search and find of density peaks," *Science*, vol. 344, no. 6191, pp. 1492–1496, Jun. 2014.
[19] H. Xie, A. Zhao, S. Huang, J. Han, S. Liu, X. Xu, X. Luo, H. Pan, Q. Du, and X. Tong, "Unsupervised hyperspectral remote sensing image clustering based on adaptive density," *IEEE Geosci. Remote Sens. Lett.*, vol. 15, no. 4, pp. 632–636, Apr. 2018.
[20] E. Elhamifar and R. Vidal, "Sparse subspace clustering: Algorithm, theory, and applications," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 35, no. 11, pp. 2765–2781, Nov. 2013.
[21] J. Xie, R. Girshick, and A. Farhadi, "Unsupervised deep embedding for clustering analysis," in *Proc. Int. Conf. Mach. Learn.*, 2016, pp. 478–487.
[22] W. Hu, T. Miyato, S. Tokui, E. Matsumoto, and M. Sugiyama, "Learning discrete representations via information maximizing self-augmented training," 2017, *arXiv:1702.08720*. [Online]. Available: https://arxiv.org/abs/1702.08720
[23] A. Krause, P. Perona, and R. G. Gomes, "Discriminative clustering by regularized information maximization," in *Proc. Adv. Neural Inf. Process. Syst.*, 2010, pp. 775–783.
[24] Y. LeCun, L. Bottou, Y. Bengio, and P. Haffner, "Gradient-based learning applied to document recognition," *Proc. IEEE*, vol. 86, no. 11, pp. 2278–2324, Nov. 1998.
[25] L. Fang, G. Liu, S. Li, P. Ghamisi, and J. A. Benediktsson, "Hyperspectral image classification with squeeze multibias network," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 3, pp. 1291–1301, Mar. 2018.
[26] J. Zhu, L. Fang, and P. Ghamisi, "Deformable convolutional neural networks for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 15, no. 8, pp. 1254–1258, Aug. 2018.
[27] Y. Tarabalka, M. Fauvel, J. Chanussot, and J. A. Benediktsson, "SVM- and MRF-based method for accurate classification of hyperspectral images," *IEEE Geosci. Remote Sens. Lett.*, vol. 7, no. 4, pp. 736–740, Oct. 2010.
[28] H. Zhang, H. Zhai, L. Zhang, and P. Li, "Spectral–spatial sparse subspace clustering for hyperspectral remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 6, pp. 3672–3684, Jun. 2016.
[29] H. Zhai, H. Zhang, L. Zhang, P. Li, and A. Plaza, "A new sparse subspace clustering algorithm for hyperspectral remote sensing imagery," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 1, pp. 43–47, Jan. 2017.
[30] H. Zhai, H. Zhang, L. Zhang, and P. Li, "Total variation regularized collaborative representation clustering with a locally adaptive dictionary for hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 1, pp. 166–180, Jan. 2018.
[31] J. M. Murphy and M. Maggioni, "Unsupervised clustering and active learning of hyperspectral images with nonlinear diffusion," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 3, pp. 1829–1845, Mar. 2018.
[32] S. Chopra, R. Hadsell, and Y. LeCun, "Learning a similarity metric discriminatively, with application to face verification," in *Proc. IEEE Comput. Soc. Conf. Comput. Vis. Pattern Recognit.*, Jun. 2005, pp. 539–546.
[33] T. Lu, S. Li, L. Fang, L. Bruzzone, and J. A. Benediktsson, "Set-to-set distance-based spectral–spatial classification of hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 12, pp. 7122–7134, Dec. 2016.
[34] L. Fang, N. He, S. Li, A. J. Plaza, and J. Plaza, "A new spatial–spectral feature extraction method for hyperspectral images using local covariance matrix representation," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 6, pp. 3534–3546, Jun. 2018.
[35] H. Cevikalp and B. Triggs, "Face recognition based on image sets," in *Proc. IEEE Comput. Soc. Conf. Comput. Vis. Pattern Recognit.*, Jun. 2010, pp. 2567–2573.
[36] J. Carreira, R. Caseiro, J. Batista, and C. Sminchisescu, "Free-form region description with second-order pooling," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 37, no. 6, pp. 1177–1189, Jun. 2015.
[37] P. Li, Q. Wang, W. Zuo, and L. Zhang, "Log-Euclidean kernels for sparse representation and dictionary learning," in *Proc. IEEE Int. Conf. Comput. Vis.*, Dec. 2013, pp. 1601–1608.
[38] Y. Pang, Y. Yuan, and X. Li, "Gabor-based region covariance matrices for face recognition," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 18, no. 7, pp. 989–993, Jul. 2008.
[39] Z. Huang, R. Wang, S. Shan, L. Van Gool, and X. Chen, "Cross Euclidean-to-Riemannian metric learning with application to face recognition from video," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 40, no. 12, pp. 2827–2840, Dec. 2018.
[40] M.-Y. Liu, O. Tuzel, S. Ramalingam, and R. Chellappa, "Entropy rate superpixel segmentation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2011, pp. 2097–2104.
[41] Y. Chen, Z. Lin, X. Zhao, G. Wang, and Y. Gu, "Deep learning-based classification of hyperspectral data," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 7, no. 6, pp. 2094–2107, Jun. 2014.
[42] Z. Zhong, J. Li, Z. Luo, and M. Chapman, "Spectral–spatial residual network for hyperspectral image classification: A 3-D deep learning framework," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 2, pp. 847–858, Feb. 2018.
[43] L. Mou, P. Ghamisi, and X. X. Zhu, "Unsupervised spectral–spatial feature learning via deep residual Conv–Deconv network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 1, pp. 391–406, Jan. 2018.
[44] M. E. Paoletti, J. M. Haut, J. Plaza, and A. Plaza, "A new deep convolutional neural network for fast hyperspectral image classification," *ISPRS J. Photogramm. Remote Sens.*, vol. 145, pp. 120–147, Nov. 2018.
[45] S. Tokui, K. Oono, S. Hido, and J. Clayton, "Chainer: A next-generation open source framework for deep learning," in *Proc. Workshop Mach. Learn. Syst. (LearningSys), 29th Annu. Conf. Neural Inf. Process. Syst. (NIPS)*, vol. 5, 2015, pp. 1–6.
[46] G. E. Hinton, N. Srivastava, A. Krizhevsky, I. Sutskever, and R. R. Salakhutdinov, "Improving neural networks by preventing co-adaptation of feature detectors," 2012, *arXiv:1207.0580*. [Online]. Available: https://arxiv.org/abs/1207.0580
[47] V. Nair and G. E. Hinton, "Rectified linear units improve restricted Boltzmann machines," in *Proc. 27th Int. Conf. Mach. Learn. (ICML)*, 2010, pp. 807–814.
[48] G. Hinton, N. Srivastava, and K. Swersky, "Lecture 6a overview of mini-batch gradient descent," Coursera Lecture Slides, 2012. [Online]. Available: https://class.coursera.org/neuralnets-2012-001/lecture
[49] J. Munkres, "Algorithms for the assignment and transportation problems," *J. Soc. Ind. Appl. Math.*, vol. 5, no. 1, pp. 32–38, 1957.



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